

SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE

SUBJECT NAME: T101 ARTIFICIAL INTELLIGENCE Practical

PRACTICAL - 3

AIM :-

To implement the A (A-Star) Search Algorithm for finding the optimal path from Churchgate to Andheri in the Mumbai Suburban Railway map.

Code :-

```
#T101 RAJDEEP M PARAB
```

```
# AIM: Implement A* Search Algorithm for Mumbai Suburban Map Problem.
```

```
dict_hn = {  
    'Churchgate': 18,  
    'Marine Lines': 17,  
    'Charni Road': 16,  
    'Grant Road': 15,  
    'Mumbai Central': 14,  
    'Mahalaxmi': 13,  
    'Lower Parel': 11,  
    'Prabhadevi': 10,  
    'Dadar': 9,  
    'Matunga Road': 8,  
    'Mahim': 7,  
    'Bandra': 6,  
    'Khar Road': 5,  
    'Santacruz': 4,  
    'Vile Parle': 2,  
    'Andheri': 0  
}  
dict_gn = dict(  
    Churchgate=dict(  
        **{'Marine Lines': 1.5}  
    ),  
    **{  
        'Marine Lines': {  
            'Churchgate': 1.5,  
            'Charni Road': 1.2  
        }  
    },  
    **{  
        'Charni Road': {  
            'Marine Lines': 1.2,  
            'Grant Road': 1.4  
        }  
    }  
)
```

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```
},
**{
  'Grant Road': {
    'Charni Road': 1.4,
    'Mumbai Central': 1.3
  }
},
**{
  'Mumbai Central': {
    'Grant Road': 1.3,
    'Mahalaxmi': 1.6
  }
},
**{
  'Mahalaxmi': {
    'Mumbai Central': 1.6,
    'Lower Parel': 1.5
  }
},
**{
  'Lower Parel': {
    'Mahalaxmi': 1.5,
    'Prabhadevi': 1.3
  }
},
**{
  'Prabhadevi': {
    'Lower Parel': 1.3,
    'Dadar': 1.2
  }
},
**{
  'Dadar': {
    'Prabhadevi': 1.2,
    'Matunga Road': 1.4
  }
},
**{
  'Matunga Road': {
    'Dadar': 1.4,
    'Mahim': 1.3
  }
}
```

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```
},
**{
    'Mahim': {
        'Matunga Road': 1.3,
        'Bandra': 2.0
    }
},
**{
    'Bandra': {
        'Mahim': 2.0,
        'Khar Road': 1.4
    }
},
**{
    'Khar Road': {
        'Bandra': 1.4,
        'Santacruz': 1.3
    }
},
**{
    'Santacruz': {
        'Khar Road': 1.3,
        'Vile Parle': 1.8
    }
},
**{
    'Vile Parle': {
        'Santacruz': 1.8,
        'Andheri': 2.0
    }
},
**{
    'Andheri': {
        'Vile Parle': 2.0
    }
}
)
import queue as Q
start = 'Churchgate'
goal = 'Andheri'
result = ""
def get_fn(citystr):
```

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```
cities = citystr.split(" , ")
hn = gn = 0
for ctr in range(0, len(cities) - 1):
    gn = gn + dict_gn[cities[ctr]][cities[ctr + 1]]
hn = dict_hn[cities[len(cities) - 1]]
return (hn + gn)
def expand(cityq):
    global result
    tot, citystr, thiscity = cityq.get()
    if thiscity == goal:
        result = citystr + " : : " + str(tot)
        return
    for cty in dict_gn[thiscity]:
        cityq.put(
            (
                get_fn(citystr + " , " + cty),
                citystr + " , " + cty,
                cty
            )
        )
    expand(cityq)
def main():
    cityq = Q.PriorityQueue()
    thiscity = start
    cityq.put(
        (
            get_fn(start),
            start,
            thiscity
        )
    )
    expand(cityq)
    print("The A* path with the total is: ")
    print(result)
main()
```

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Output :-

```
IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/rajde/OneDrive/Rajdeep's File/AI/AI_Practical-3/Practical-3.py
The A* path with the total is:
Churchgate , Marine Lines , Charni Road , Grant Road , Mumbai Central , Mahalaxmi , Lower Parel , Prabhadevi , Dadar , Matunga Ro
ad , Mahim , Bandra , Khar Road , Santacruz , Vile Parle , Andheri : : 22.200000000000003
>>>
===== RESTART: C:/Users/rajde/OneDrive/Rajdeep's File/AI/AI_Practical-3/Practical-3.py =====
The A* path with the total is:
Churchgate , Marine Lines , Charni Road , Grant Road , Mumbai Central , Mahalaxmi , Lower Parel , Prabhadevi , Dadar , Matunga Ro
ad , Mahim , Bandra , Khar Road , Santacruz , Vile Parle , Andheri : : 22.200000000000003
>>>
```